

Algorithms and data Structures - Lab Sheet 1 (37607)

Question 01)

```
#Traverse

arr = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

print("Arrayelements:")
for i in range(len(arr)):
    print(arr[i])
```

Output -

```
===== REST
Arrayelements:
10
20
30
40
50
60
70
80
90
100
```

Question 02)

```
arr = [10, 20, 30, 40, 50]

value = int(input("Enter the element to insert: "))
index = int(input("Enter the index: "))

arr.insert(index, value)

print("Updated array:")
print(arr)
```

Output-

```
===== REST =====
Enter the element to insert: 2
Enter the index: 3
Updated array:
[10, 20, 30, 2, 40, 50]
```

Question 03)

```
array = [10,20,30,40,50]

index = int(input("enter the index of the element you want to delete: "))

del array[index]
print(array)
```

Output –

```
enter the index of the element you want to delete: 4
[10, 20, 30, 40]
```

Question 04)

```
arr = [10,20,30,40,50]

a = int(input("Enter index"))
new = int(input("Enter new value"))
arr[2] = new
print("Updated array")
print(arr)
```

Output –

```
Enter index2
Enter new value25
Updated array
[10, 20, 25, 40, 50]
```

Question 05)

```
# Input size and elements of first array
size1 = int(input("Enter the size of the first array: "))
arr1 = []

for i in range(size1):
    element = int(input("Enter element: "))
    arr1.append(element)

# Input size and elements of second array
size2 = int(input("Enter the size of the second array: "))
arr2 = []

for i in range(size2):
    element = int(input("Enter element: "))
    arr2.append(element)

# Merge the arrays into a third array
arr3 = arr1 + arr2

# Display the merged array
print("Merged array:")
print(arr3)
```

Output –

```
Enter the size of the first array: 5
Enter element: 10
Enter element: 20
Enter element: 30
Enter element: 40
Enter element: 50
Enter the size of the second array: 4
Enter element: 1
Enter element: 2
Enter element: 3
Enter element: 4
Merged array:
[10, 20, 30, 40, 50, 1, 2, 3, 4]
```